

From Single-Asset Timing to Scenario-Based Portfolio Allocation

A Four-Phase Study of Backtest Design, Parameter Sensitivity, Mean–Variance Allocation, and CVaR

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Abstract

This report documents four stages in the development of a small Python backtesting and portfolio-allocation engine. Phase 1 implemented a day-by-day moving-average strategy with a fixed 5% stop and separated signal formation from the return subsequently earned. Phase 2 replaced the common percentage stop with an entry-time Average True Range (ATR) distance. The change did not improve every asset: under the AAPL-selected parameters, the reported 2025 return was unchanged for AAPL, decreased from -5.4% to -6.5% for TSLA, and increased from 8.9% to 13.7% for BTC-USD. This mixed result motivated a change in question from single-asset timing to cross-asset allocation.

Phase 3 estimated a long-only Markowitz model on 2025 daily returns for AAPL, TSLA, BTC-USD, and GLD. The maximum-Sharpe solution allocated 89.6% to GLD and achieved an in-sample Sharpe ratio of 2.65. With those weights frozen, a single January–June 2026 holdout returned -2.12% with a Sharpe ratio of 0.01, compared with -7.30% and -0.56 for equal weight. Phase 4 replaced variance with scenario CVaR, using fitted Student- t marginals and a Gaussian copula to generate 20,000 joint daily-return scenarios. The 90% CVaR portfolio returned -3.30% in the same holdout and reduced maximum drawdown and worst-day loss relative to the Phase 3 portfolio, but it did not dominate either Phase 3 or equal weight on every metric. The study therefore supports a narrower conclusion: changing the risk objective changes which losses are emphasized, but one fixed holdout cannot establish persistent alpha or identify the contribution of any single modelling choice.

1 Research Question and Version Design

The project began with a practical engineering question: how can a backtest make decisions without allowing information from day t to leak into a signal that is meant to exist at day $t - 1$? Once the event loop was made explicit, the main difficulty shifted. A rule that appeared reasonable on one asset was not automatically portable to another, and a portfolio optimized on one period did not retain its in-sample performance in the next.

The four phases are therefore not four claims of improvement. They are four changes in the object being studied:

Table 1. Version sequence and the question addressed at each stage.

Phase	Implementation	Question
V1	MA crossover and fixed 5% stop	Can a close-to-close event loop avoid forward signal leakage, and how does one common stop distance behave across assets?
V2	Entry-time $k \times \text{ATR}$ stop	Does scaling the stop by recent price range transfer the AAPL-selected rule to TSLA and BTC-USD?
V3/V3.1	Markowitz allocation and fixed holdout	What changes when the decision variable becomes portfolio weight, and how stable are 2025 weights in early 2026?
V4	Student- t scenarios and CVaR minimization	How do allocations change when optimization targets simulated tail loss rather than variance or Sharpe ratio?

2 Data and Evaluation Conventions

Phases 1 and 2 downloaded 2025 OHLC data separately for AAPL, TSLA, and BTC-USD. AAPL was used for parameter selection, and the selected parameters were then applied unchanged to the other two assets. A 0.1% commission was deducted on entries and exits. Sharpe ratios used 252 annualization days for equities and 365 for BTC-USD. Because parameter selection and reported AAPL performance use the same period, the AAPL result is in-sample.

Phases 3 and 4 used the cached file `v3_market_data_2025.csv`. It contains 364 dated rows from January 1 to December 30, 2025. Requiring prices for all four assets leaves 249 shared trading dates and 248 simple daily-return observations. This common-date rule removes BTC weekend observations rather than treating closed equity markets as zero-return days. Expected returns were annualized as $252\bar{r}$, and the sample covariance matrix as 252Σ . The risk-free rate was set to zero.

The fixed out-of-sample window runs from January 1 to June 15, 2026. V3.1 and V4 downloaded that window at run time and applied fixed portfolio weights to each day’s asset returns. This calculation corresponds to a constant-weight portfolio rebalanced each day, without trading costs. The window is one holdout comparison, not a repeated or rolling validation design.

3 Phase 1: Event Loop and Fixed Percentage Stop

3.1 Signal Timing

For each asset, the strategy formed a Boolean signal from a short and long moving average. A new position was opened at the previous close when the signal changed from false to true, so that the next daily return was earned only after the signal existed. This indexing choice prevents the crossover calculated at the end of day t from being credited with the return already realized during day t .

The fixed-stop implementation is more approximate. It checks whether the current close is at or below 95% of the entry price and, if so, records an exit at the theoretical stop price. It therefore does not model intraday paths, bid–ask spread, or a gap through the stop. The assumption is retained as part of the early engine, but its output should be read as an experiment with rule behavior rather than executable performance.

3.2 2025 Results

The AAPL grid search selected MA(8, 60). Applying that pair and a 5% stop unchanged produced the results in Table 2.

Table 2. Phase 1 reported performance under MA(8, 60) and a fixed 5% stop.

Asset	Return	Sharpe	Trades
AAPL	31.6%	2.16	1
TSLA	-5.4%	-0.05	4
BTC-USD	8.9%	0.58	3



Figure 1. Phase 1 AAPL path under the selected MA(8, 60) rule and fixed stop.

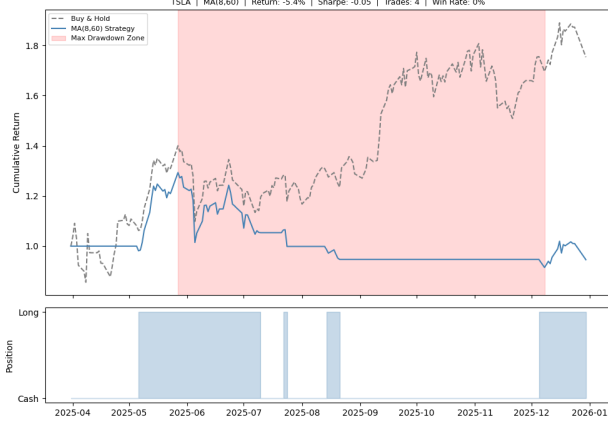


Figure 2. Phase 1 TSLA path.

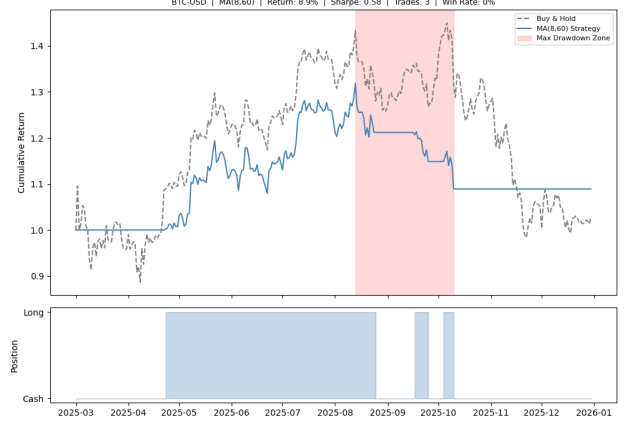


Figure 3. Phase 1 BTC-USD path.

The common parameters did not produce a common performance profile. AAPL had one long holding period and a positive return. TSLA repeatedly moved between cash and a long position and finished slightly negative despite a much stronger buy-and-hold path. BTC-USD remained positive but well below its mid-year peak. These outcomes do not isolate the stop as the only cause: the moving-average signal, the stop rule, and each asset path operate together. They do show that a percentage threshold selected on AAPL cannot be treated as a volatility-neutral control.

4 Phase 2: ATR Scaling and Non-Portable Calibration

Phase 2 replaced the 5% distance with a 14-day Average True Range. For day t , true range was

$$TR_t = \max \{H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|\},$$

and ATR was the 14-day rolling mean of TR_t . At entry, the stop was set to

$$S_{\text{entry}} = P_{\text{entry}} - k \text{ ATR}_{t-1}.$$

This is an ATR-scaled stop fixed at entry, not a trailing stop. A three-way grid search over short MA, long MA, and k selected MA(8, 60) and $k = 1.5$ on AAPL.

Table 3. Phase 1 and Phase 2 reported outcomes under their AAPL-selected settings.

Asset	Stop rule	Return	Sharpe	Trades
AAPL	Fixed 5%	31.6%	2.16	1
AAPL	$1.5 \times \text{ATR}$	31.6%	2.16	1
TSLA	Fixed 5%	-5.4%	-0.05	4
TSLA	$1.5 \times \text{ATR}$	-6.5%	-0.09	4
BTC-USD	Fixed 5%	8.9%	0.58	3
BTC-USD	$1.5 \times \text{ATR}$	13.7%	0.84	3

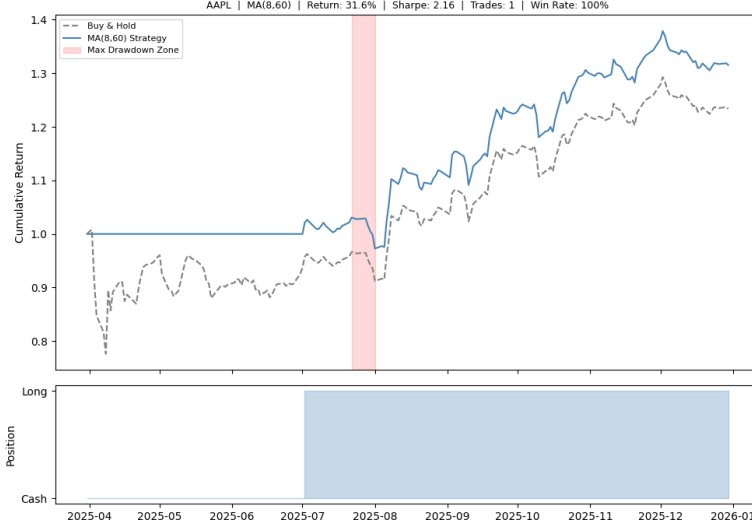


Figure 4. Phase 2 AAPL path under MA(8, 60) and a $1.5 \times \text{ATR}$ entry stop.

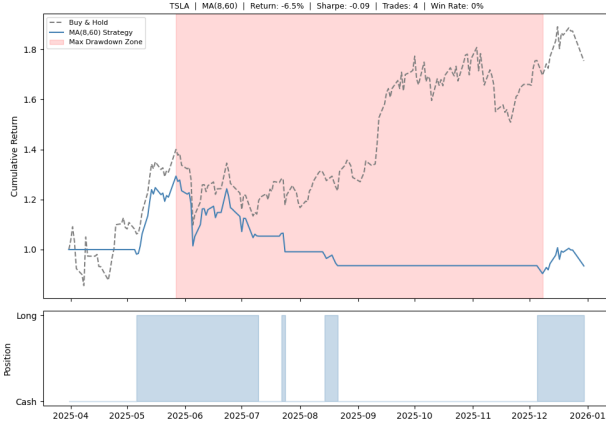


Figure 5. Phase 2 TSLA path.

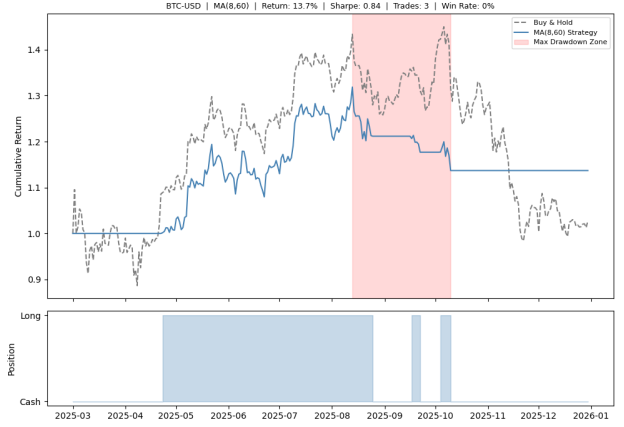


Figure 6. Phase 2 BTC-USD path.

The change was not uniformly beneficial or uniformly harmful. It left AAPL's headline result unchanged, slightly worsened TSLA, and improved BTC-USD. This is more informative than describing Phase 2 as a successful repair or a complete failure. ATR supplies a price-scaled unit, but the multiplier, moving-average pair, entry logic, and market path remain jointly calibrated. Selecting all three on one AAPL year did not establish portability.

5 Phase 3: Mean–Variance Portfolio Allocation

Phase 3 changed the decision from when to hold one asset to how to allocate capital across four assets. Let w denote long-only portfolio weights with $\sum_i w_i = 1$, μ annualized sample mean returns, and Σ the annualized sample covariance matrix. The two objectives were

$$\min_w \sqrt{w^\top \Sigma w} \quad \text{and} \quad \max_w \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}},$$

corresponding to global minimum variance and maximum Sharpe ratio. This is the single-period mean–variance framework introduced by Markowitz [1]. Both problems were solved with SciPy's SLSQP implementation under long-only bounds [3].

5.1 Correlation Structure

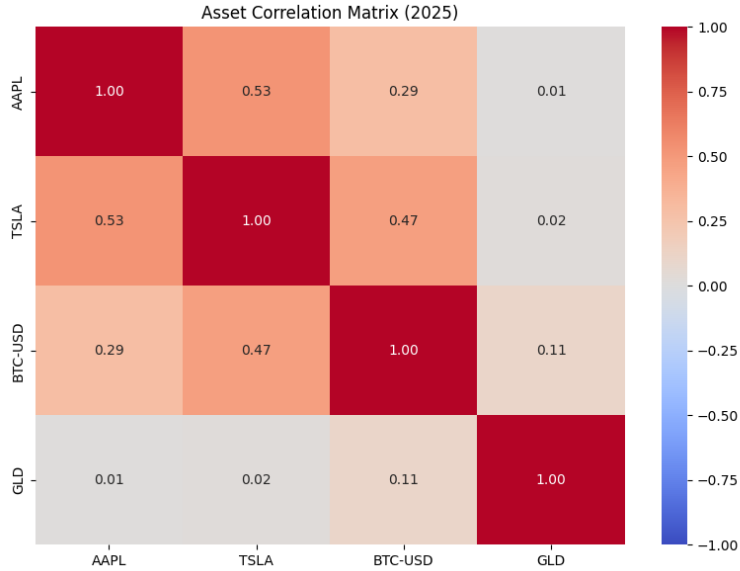


Figure 7. Phase 3 Pearson correlations from 248 daily-return observations on shared 2025 trading dates.

GLD had correlations of 0.012 with AAPL, 0.023 with TSLA, and 0.105 with BTC-USD. Those values were lower than the correlations among AAPL, TSLA, and BTC-USD. The result explains why both optimized portfolios assigned a large weight to GLD, but it does not imply that the correlation estimates are stable outside 2025.

5.2 Optimized Weights and In-Sample Metrics

Table 4. Phase 3 optimized allocations and in-sample annualized metrics.

Portfolio	AAPL	TSLA	BTC	GLD	Return	Volatility	Sharpe
GMV	22.9%	0.0%	8.7%	68.5%	38.98%	16.73%	2.33
Max Sharpe	6.2%	4.3%	0.0%	89.6%	48.71%	18.36%	2.65

The large annualized returns reflect the arithmetic scaling of one unusually strong training year; they are not forecasts with uncertainty intervals. The concentration in GLD is itself evidence of estimation sensitivity rather than proof of a durable optimal mix.

5.3 Random-Portfolio Sanity Check

Ten thousand random long-only weight vectors were generated using a fixed seed and plotted against the two SLSQP solutions.

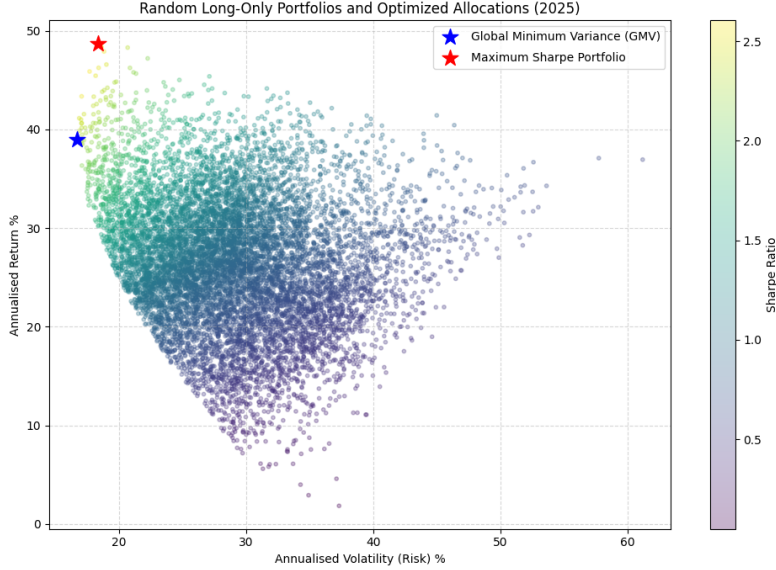


Figure 8. Random feasible portfolios and the Phase 3 optimizer solutions. The point cloud is a visual numerical check, not a proof of global optimality.

The optimized points lie on the favorable boundary of the sampled cloud. This checks that the returned solutions are consistent with 10,000 randomly drawn feasible portfolios. Random search cannot prove convergence because it does not exhaust the continuous weight simplex.

5.4 V3.1: Fixed 2026 Holdout

V3.1 froze the full-precision 2025 maximum-Sharpe weights and applied them to the fixed early-2026 window.

Table 5. V3.1 fixed holdout comparison using the download interval from January 1 up to, but not including, June 15, 2026.

Portfolio	Cumulative return	Annualized Sharpe
V3 maximum Sharpe	-2.12%	0.01
Equal weight $1/N$	-7.30%	-0.56

The optimized portfolio lost less than equal weight in this window, but its in-sample Sharpe ratio did not persist. The comparison does not establish alpha, and it cannot identify whether the difference came from expected-return estimation, covariance, or the realized path of GLD.

6 Phase 4: Student- t Scenarios and CVaR

6.1 Scenario Construction

Variance treats upside and downside dispersion symmetrically and relies on sample moments. Phase 4 instead fitted a univariate Student- t distribution to each asset’s 2025 daily returns and generated joint scenarios through a Gaussian copula. If R is the Pearson correlation matrix and $LL^\top = R$, the simulation was

$$z = L\varepsilon, \quad u_i = \Phi(z_i), \quad \tilde{r}_i = F_{t,i}^{-1}(u_i),$$

where $\varepsilon \sim N(0, I)$, Φ is the standard normal CDF, and $F_{t,i}$ is the fitted Student- t marginal for asset i . A guard would refit a marginal with three degrees of freedom if the unconstrained estimate were at most two, ensuring finite variance. No guard message appeared in the reported run, so this condition was not activated.

Using seed 42, the code produced 20,000 joint one-day scenarios. The Gaussian copula preserves a rank transformation of linear dependence while the Student- t marginals allow heavier univariate tails. It does not add the tail dependence of a multivariate t -copula.

6.2 CVaR Objective

For scenario return matrix $\tilde{R}$, portfolio loss in scenario s was $\ell_s(w) = -\tilde{R}_s w$. The implemented empirical objective was the mean loss among the worst $1 - \alpha$ fraction:

$$\widehat{\text{CVaR}}_\alpha(w) = \frac{1}{|\mathcal{T}_\alpha(w)|} \sum_{s \in \mathcal{T}_\alpha(w)} \ell_s(w).$$

This is the tail-loss concept formalized in the CVaR optimization literature by Rockafellar and Uryasev [2]. The code minimized the sorted-tail objective with SLSQP for $\alpha \in \{0.90, 0.95, 0.99\}$, subject to the same long-only, fully invested constraint. All three solver calls reported success.

Table 6. Phase 4 optimized scenario CVaR objectives and allocations.

α	Scenario CVaR	AAPL	TSLA	BTC-USD	GLD
0.90	1.7694%	15.29%	0.00%	9.21%	75.50%
0.95	2.2740%	11.80%	0.43%	12.86%	74.91%
0.99	3.5355%	5.36%	2.40%	19.81%	72.42%

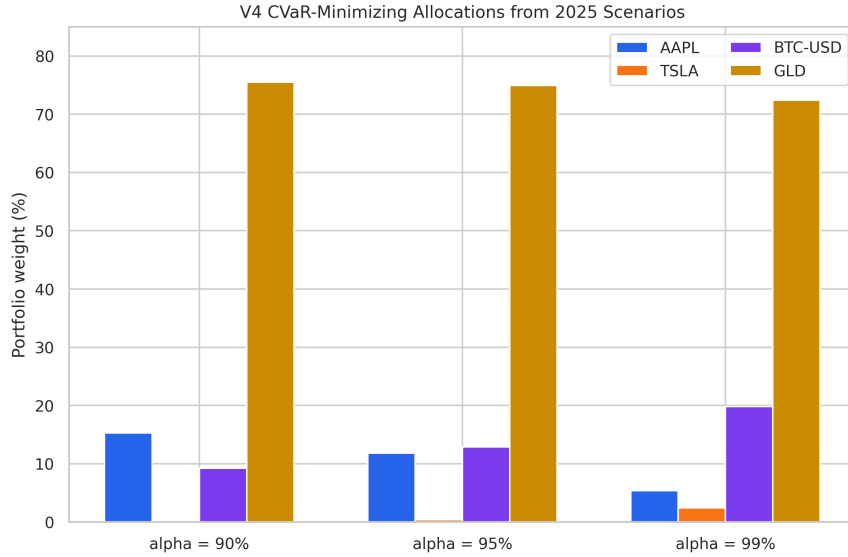


Figure 9. Phase 4 long-only allocations across the three CVaR confidence levels.

Higher α did not mechanically produce a more GLD-heavy allocation. In these fitted scenarios, the BTC-USD weight increased from 9.21% to 19.81% while GLD remained near 72–76%. A higher confidence level changes which simulated losses enter the objective; it does not impose an ordering on individual asset weights.

6.3 Fixed 2026 Holdout Comparison

The V4 portfolios were assessed on the same window as V3.1. V4 also recomputed the V3 benchmark using weights rounded to four decimal places; the rounding is small but means the benchmark is an approximation to the full-precision V3.1 weights.

Table 7. Fixed 2026 holdout metrics reported by Phase 4.

Portfolio	Return	Sharpe	Max drawdown	Worst day
CVaR 90%	-3.30%	-0.13	-19.41%	-7.73%
CVaR 95%	-4.68%	-0.24	-20.70%	-7.69%
CVaR 99%	-7.30%	-0.43	-22.92%	-7.44%
V3 maximum Sharpe	-2.12%	0.01	-21.66%	-9.03%
Equal weight	-7.30%	-0.56	-15.13%	-4.79%

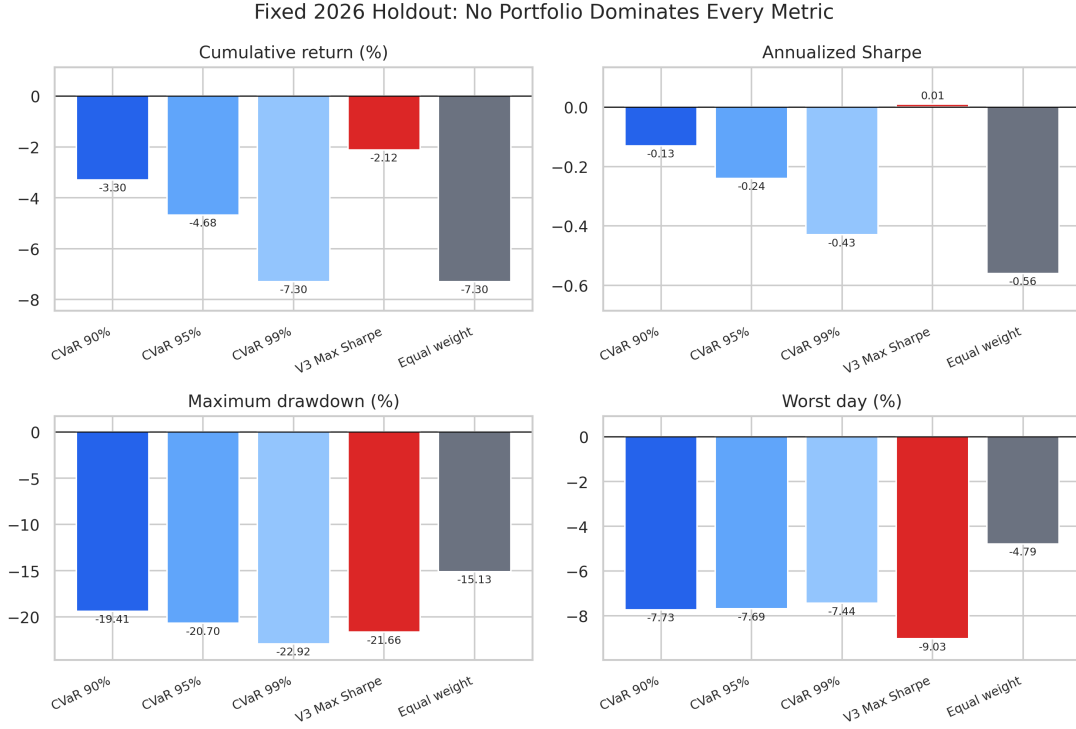


Figure 10. The fixed holdout produces a metric trade-off rather than one dominant portfolio.

The V3 portfolio had the least negative cumulative return and the highest Sharpe ratio. The 90% and 95% CVaR portfolios improved maximum drawdown and worst-day loss relative to V3, but equal weight had the shallowest drawdown and mildest worst day. Within the CVaR group, increasing α improved the worst day slightly while cumulative return and maximum drawdown worsened. These rankings are descriptive outcomes from one realized path. They do not show that 90% is generally optimal or that 99% CVaR necessarily overfits.

7 What the Four Phases Establish

Three conclusions are supported by the implemented comparisons.

1. **Timing correctness is necessary but not sufficient.** Delaying signal use until the following return removes one form of look-ahead bias, but fill assumptions and parameter selection still shape the result.
2. **Scaling a rule is not the same as validating it.** ATR gives the stop a volatility-dependent unit, yet the AAPL-selected specification produced mixed cross-asset changes. The experiment did not separate the multiplier from the moving-average and path effects.
3. **Risk objectives select different trade-offs.** Mean–variance and scenario CVaR emphasized different features of the same four-asset training data. In the holdout, no method dominated return, Sharpe, drawdown, and worst-day loss simultaneously.

The transition from V3 to V4 changed several modelling choices at once: univariate marginal distributions, scenario generation, and the objective function. The comparison therefore cannot attribute the V4 result to Student- t fitting, the Gaussian copula, or CVaR individually.

8 Limitations and Reproducibility

- **Execution.** V1 and V2 use close-based stop checks and theoretical fills. They omit intraday paths, spreads, slippage, and gap risk. Their trade log records the exit-day strategy increment rather than full trade P&L, so win-rate fields are not used in this report.
- **Selection.** V1 and V2 tune on AAPL and report the same AAPL period. V3 and V4 estimate parameters from one calendar year. None of the phases performs rolling or nested model selection.
- **Portfolio mechanics.** V3.1 and V4 apply constant weights to each day's returns, equivalent to costless daily rebalancing. A buy-and- hold portfolio with drifting weights would follow a different path.
- **Distribution.** Student- t marginals allow heavier univariate tails, but the Gaussian copula does not model non-zero asymptotic tail dependence. The Pearson correlation matrix is also treated as fixed.
- **Optimization.** The V4 sorted-tail objective is convex but non-smooth at scenario-order boundaries. SLSQP returned successful status, but no linear-program formulation, multiple-start check, or simulation- size sensitivity analysis was used. The V3 random cloud is a visual check, not a proof.
- **Data versioning.** The 2025 training prices are cached and reproduce the reported V3 correlation matrix and weights. The 2026 holdout was downloaded live and is not yet cached, so future data-source revisions could prevent byte-for-byte reproduction of the OOS metrics.

9 Conclusion

The project evolved by changing the question whenever the previous version could not support the interpretation being placed on it. V1 made the event order explicit. V2 showed that a volatility-scaled stop did not create a universally transferable rule. V3 moved from directional timing to covariance-based allocation, then V3.1 showed how sharply its in-sample Sharpe ratio weakened in a fixed holdout. V4 changed the target from variance and expected return to simulated tail loss, but its OOS advantages depended on the metric.

The strongest result is therefore not a winning portfolio. It is a clearer separation among implementation correctness, in-sample optimization, and out-of-sample evidence. Each can be computed correctly while answering a different statistical question.

Code and Data Map

File	Role
<code>backtest.py</code>	V1 moving-average engine and fixed stop
<code>backtest_v2_atr_overfit_experiment.py</code>	V2 ATR-scaled entry stop and cross-asset application
<code>backtest_v3_markowitz_portfolio.py</code>	V3 correlation, GMV, maximum Sharpe, and random portfolios
<code>backtest_v3_1_oos_forward_test.py</code>	V3.1 fixed early-2026 holdout
<code>backtest_v4_cvar_scenario_allocation.py</code>	V4 Student- t scenario generation, CVaR allocation, and OOS comparison
<code>v3_market_data_2025.csv</code>	Cached 2025 Close series used by V3 and V4

References

- [1] H. Markowitz. Portfolio Selection. *The Journal of Finance*, 7(1):77–91, 1952. <https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>.
- [2] R. T. Rockafellar and S. Uryasev. Optimization of Conditional Value-at-Risk. *The Journal of Risk*, 2(3):21–41, 2000. <https://sites.math.washington.edu/~rtr/papers/rtr179-CVaR1.pdf>.
- [3] SciPy Developers. `scipy.optimize.minimize`: SLSQP documentation. <https://docs.scipy.org/doc/scipy/reference/optimize.minimize-slsqp.html>.